SMARTBRIDGE Let's Bridge the Gap

SKILL WALLET A SmartBridge Product

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Name	Rising waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to develop an intelligent flood prediction system using machine learning techniques. The system analyses historical weather data, including annual rainfall, cloud visibility, and seasonal rainfall patterns, to predict the likelihood of flood events. Multiple classification algorithms such as Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost are trained and evaluated to identify the best-performing model. The selected model is integrated into a Flask web application, providing an easy-to-use interface for flood risk prediction.

Project Overview	
Objective	To develop a machine learning-based flood prediction system that accurately predicts flood risk using historical weather data and provides an early warning solution through a Flask web application ..
Scope	The project focuses on analysing rainfall and weather-related parameters to predict flood occurrence. It includes data preprocessing, model training, performance evaluation, and deployment of the bestperforming model using Flask for real-time prediction.
Problem Statement	

Description	Floods cause significant damage to life and property every year. Traditional flood forecasting methods may not provide timely and accurate predictions, limiting the effectiveness of disaster preparedness and emergency response
Impact	The project improves early flood warning capabilities, supports better resource allocation, reduces disaster response time, and enhances public safety in flood-prone regions.
Proposed Solution	
Approach	The solution applies supervised machine learning algorithms including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost. The models are trained using historical weather data, and the best-performing model is integrated into a Flask web application for real-time flood prediction.
Key Features	• Historical weather data analysis • Flood risk prediction using machine learning • Comparison of multiple classification algorithms • Flask-based web application • User-friendly prediction interface • Early warning support for disaster management

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU	Intel Core i5 or above

Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	20 GB Free Space
Software		
Framework	Web framework	Flask
Libraries	Python libraries	Pandas, NumPy, Scikit-learn, XGBoost, Joblib
Development Environment	IDE	Jupyter Notebook, Visual Studio
Data		
Data	Historical Weather Data	CSV file containing rainfall, cloud visibility, seasonal rainfall and flood labels